

We already discussed the following theorem and in what follows, we will be building on it.

Theorem (Matrix Inverses)

Let A be an $n \times n$ matrix. Then the following conditions are equivalent.

- ① A is invertible.
- ② the reduced row-echelon form on A is I .
- ③ $[A \mid I_n]$ can be transformed into $[I_n \mid A^{-1}]$ using the Matrix Inversion Algorithm.

The Invertible Matrix Theorem

Theorem

Let A be an $n \times n$ matrix, and let $\vec{X}$, $\vec{B}$ be $n \times 1$ vectors. The following conditions are equivalent.

- 1 A is invertible.
- 2 $A\vec{X} = \vec{0}$ has only the trivial solution, $\vec{X} = \vec{0}$. ($T_A(\vec{x}) = A\vec{x}$ is one-to-one.)
- 3 A can be transformed to I_n by elementary row operations.
- 4 A has n pivot positions.
- 5 The columns of A are linearly independent.
- 6 The columns of A span $\mathbb{R}^n$. ($T_A(\vec{x}) = A\vec{x}$ is onto.)
- 7 The system $A\vec{X} = \vec{B}$ has a unique solution $\vec{X}$ for any choice of $\vec{B}$.
- 8 There exists an $n \times n$ matrix C with the property that $AC = I_n$.
- 9 There exists an $n \times n$ matrix D with the property that $DA = I_n$.
- 10 A^T is invertible.

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We already discussed if 1, then the RREF of A will be I_n , which means $A\vec{x} = \vec{0}$ implies $\vec{x} = \vec{0}$ (2). Also, if 2, since the matrix A is $n \times n$, it means in every row there is a leading one after row reduction. Hence, its RREF is I_n and therefore A is invertible.

$\boxed{1 \longleftrightarrow 3}$: We already discussed this equivalence.

$\boxed{1 \longleftrightarrow 4}$: We have also already discussed this equivalence. Note that the RREM being I_n is the same as having n pivot positions.

$\boxed{2 \longleftrightarrow 5}$: We proved a few sessions ago that columns of A are linearly independent iff $A\vec{x} = \vec{0}$ implies $\vec{x} = \vec{0}$.

$\boxed{3 \longleftrightarrow 6}$: If A can be transformed to I_n , then $A\vec{x} = \vec{b}$ is a consistent SLE for all $\vec{b} \in \mathbb{R}^n$ and hence columns of A span $\mathbb{R}^n$, i.e., for all $\vec{b}$ in $\mathbb{R}^n$ we can find a linear combination $x_1\vec{A}_1 + x_2\vec{A}_2 + \dots + x_n\vec{A}_n = \vec{b}$. Also, if columns of A span $\mathbb{R}^n$, then $A\vec{x} = \vec{b}$ is a consistent SLE for all $\vec{b} \in \mathbb{R}^n$ and the last row of REM cannot be all zeros, which means there are n leading ones and hence the RREM is I_n .

$\longleftrightarrow$

$\boxed{3 \longleftrightarrow 7}$: If A can be reduced to I_n , then we know that the solution to $A\vec{x} = \vec{b}$ exists uniquely for all $\vec{b} \in \mathbb{R}^n$. Also, if $A\vec{x} = \vec{b}$ has a unique solution for any $\vec{b} \in \mathbb{R}^n$, then the RREM cannot have a row of zeros, which means I_n is the RREM.

1 $\iff$ 8

: If A is invertible, by definition, we have $A\bar{A}^{-1} = I$ ($C = \bar{A}^{-1}$).

Let us assume There exists an $n \times n$ matrix C with the property that $AC = I_n$.

We can show that A is onto and hence invertible:

For any $\vec{b} \in \mathbb{R}^n$, we can write: $\vec{b} = I \vec{b} = (AC) \vec{b} = A(C\vec{b})$
 or $T_A(C\vec{b}) = \vec{b}$, which means for any $\vec{b} \in \mathbb{R}^n$, There exists $\vec{x} = C\vec{b} \in \mathbb{R}^n$ such that $T_A(\vec{x}) = \vec{b}$ and hence T is onto and hence A is invertible. (1 $\iff$ 6)

1 $\iff$ 9

: If A is invertible, by definition, we have $\bar{A}A = I$ ($D = \bar{A}$).

Now assume $DA = I$, we can show that T_A is one-to-one and hence A is invertible.

If $T_A(\vec{x}_1) = T_A(\vec{x}_2)$, then $A\vec{x}_1 = A\vec{x}_2$ and $DA\vec{x}_1 = DA\vec{x}_2$
 which give $\vec{x}_1 = \vec{x}_2$ (Since $DA = I$). Hence T_A is injective and A is invertible.

1 $\iff$ 10

If A is invertible: $(A\bar{A}^{-1})^T = I^T \implies (\bar{A}^{-1})^T A^T = I$
 $(\bar{A}^{-1}A)^T = I^T \implies A^T (\bar{A}^{-1})^T = I \implies (A^T)^{-1} = (\bar{A}^{-1})^T$

If A^T is invertible: $(A^T)^{-1} A^T = I \xrightarrow{T} A ((A^T)^{-1})^T = I$
 $A^T (A^T)^{-1} = I \xrightarrow{T} ((A^T)^{-1})^T A = I \implies ((A^T)^{-1})^T = \bar{A}^{-1}$

Example: Is the following matrix invertible?

$$\begin{array}{cccc} \vec{u}_1 & \vec{u}_2 & \vec{u}_3 & \vec{u}_4 \\ \begin{bmatrix} 1 & -1 & 1 & 2 \\ 0 & 1 & 5 & -1 \\ 1 & 0 & 7 & 1 \\ 0 & 2 & 10 & -2 \end{bmatrix} \end{array}$$

Since $\vec{u}_1 - \vec{u}_2 = \vec{u}_4$, we know that the columns are not linearly independent and hence the matrix is not invertible.

True or False?

(A) Invertible transformations map the zero vector to itself.

True

(2) of the invertible matrix theorem says that for invertible matrices we have: $A\vec{x} = \vec{0}$, implies $\vec{x} = \vec{0}$. This means $T_A(\vec{0}) = \vec{0}$.

(B) Non-invertible matrices cannot be written as a product of elementary matrices.

True

If we assume that a non-invertible matrix A can be written as a product of E.M.s ($A = E_1 E_2 \dots E_n$), we can easily see that the product is invertible since each E.M. is invertible $(E_1 E_2 \dots E_n)^{-1} = E_n^{-1} E_{n-1}^{-1} \dots E_2^{-1} E_1^{-1}$, which means we reach a contradiction. Hence, non-invertible matrices cannot be written as a product of E.M.s.

Example: Suppose P is a square matrix, which satisfies the following:

$$P^3 + 2P^2 + 2P - 3I = 0.$$

Find P^{-1} , if possible.

$$\begin{aligned} \frac{1}{3}(P^3 + 2P^2 + 2P - 3I) &= 0 \xrightarrow{\div 3} \frac{1}{3}(P^3 + 2P^2 + 2P) - I = 0 \\ \frac{1}{3}(P^3 + 2P^2 + 2P) &= I \longrightarrow P \left[\frac{1}{3}(P^2 + 2P + 2I) \right] = I \longrightarrow P^{-1} = \frac{1}{3}(P^2 + 2P + 2I) \end{aligned}$$